Perturbations in Celestial Mechanics

Ville Räisänen*

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1 Planetary Disturbing Function

1.1 Two-Planet Solar System

Consider a solar system with two planets orbiting a Sun. The position vectors and masses of the two planets and the Sun are denoted with $\boldsymbol{R}$, $\boldsymbol{R}'$, $\boldsymbol{R}_s$ and m, m', M, respectively. Then, from Newton's laws of motion, we obtain

$$\ddot{\boldsymbol{R}}_s = Gm\frac{\boldsymbol{R}-\boldsymbol{R}_s}{|\boldsymbol{R}-\boldsymbol{R}_s|^3} + Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}_s}{|\boldsymbol{R}'-\boldsymbol{R}_s|^3} \tag{1}$$

$$\ddot{\boldsymbol{R}} = Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}}{|\boldsymbol{R}'-\boldsymbol{R}|^3} + GM\frac{\boldsymbol{R}_s-\boldsymbol{R}}{|\boldsymbol{R}_s-\boldsymbol{R}|^3} \tag{2}$$

$$\ddot{\boldsymbol{R}}' = Gm\frac{\boldsymbol{R}-\boldsymbol{R}'}{|\boldsymbol{R}-\boldsymbol{R}'|^3} + GM\frac{\boldsymbol{R}_s-\boldsymbol{R}'}{|\boldsymbol{R}_s-\boldsymbol{R}'|^3}, \tag{3}$$

where G is the gravitational constant. Definition of position vectors $\boldsymbol{r} := \boldsymbol{R} - \boldsymbol{R}_s$ and $\boldsymbol{r}' := \boldsymbol{R}' - \boldsymbol{R}_s$ relative to the position vector of the Sun, yields

$$\ddot{\boldsymbol{r}} = \ddot{\boldsymbol{R}} - \ddot{\boldsymbol{R}}_s \tag{4}$$

$$= Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}}{|\boldsymbol{R}'-\boldsymbol{R}|^3} + GM\frac{\boldsymbol{R}_s-\boldsymbol{R}}{|\boldsymbol{R}_s-\boldsymbol{R}|^3} - Gm\frac{\boldsymbol{R}-\boldsymbol{R}_s}{|\boldsymbol{R}-\boldsymbol{R}_s|^3} - Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}_s}{|\boldsymbol{R}'-\boldsymbol{R}_s|^3} \tag{5}$$

$$= Gm'\left(\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} - \frac{\boldsymbol{r}'}{r'^3}\right) - \mu\frac{\boldsymbol{r}}{r^3}, \tag{6}$$

where we have defined the **standard gravitational parameter** $\mu := G(M+m)$. Reorganizing and performing the same derivation for $\ddot{\boldsymbol{r}}'$, we obtain the equations for the **three-body problem**

$$\ddot{\boldsymbol{r}} + \mu\frac{\boldsymbol{r}}{r^3} = Gm'\left(\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} - \frac{\boldsymbol{r}'}{r'^3}\right) \tag{7}$$

$$\ddot{\boldsymbol{r}}' + \mu'\frac{\boldsymbol{r}'}{r'^3} = Gm\left(\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} - \frac{\boldsymbol{r}}{r^3}\right), \tag{8}$$

*https://www.github.com/vsr83

where the left-hand side represents a two-body problem and the right-hand a **perturbation** from a third body. When the mass M is significantly greater than the masses m and m' and the distances between the two smaller masses are large, the perturbation stays small.

1.1.1 Disturbing Functions

We represent the perturbations as the gradients $\nabla\mathcal{R}$ and $\nabla\mathcal{R}'$ of the **disturbing functions** $\mathcal{R}$ and $\mathcal{R}'$. That is,

$$\nabla\mathcal{R} = Gm'\left(\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3}-\frac{\boldsymbol{r}'}{r'^3}\right), \tag{9}$$

$$\nabla'\mathcal{R}' = Gm\left(\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}-\frac{\boldsymbol{r}}{r^3}\right), \tag{10}$$

where ∇' represents gradient w.r.t. primed coordinates $(\partial/\partial x', \partial/\partial y', \partial/\partial z')$. To obtain an expression for the disturbing functions, note that

$$\frac{\partial}{\partial x}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{\partial}{\partial x}\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}} = -\frac{x-x'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}. \tag{11}$$

Thus, obviously also

$$\nabla\left(Gm'\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\right) = -Gm'\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} = Gm'\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3}, \tag{12}$$

$$\nabla'\left(Gm\frac{1}{|\boldsymbol{r}'-\boldsymbol{r}|}\right) = -Gm\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} = Gm\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}. \tag{13}$$

Similarly, we can note that

$$\frac{\partial}{\partial x}\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3} = \frac{x'}{r'^3} \quad\Rightarrow\quad \nabla\left(\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3}\right) = \frac{\boldsymbol{r}'}{r'^3}. \tag{14}$$

to obtain expressions for the disturbing function

$$\mathcal{R} = \overline{\mu}'\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}-\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3}\right) \tag{15}$$

$$\mathcal{R}' = \overline{\mu}\left(\frac{1}{|\boldsymbol{r}'-\boldsymbol{r}|}-\frac{\boldsymbol{r}'\cdot\boldsymbol{r}'}{r^3}\right), \tag{16}$$

where we have defined $\overline{\mu} := Gm$ and $\overline{\mu}' := Gm'$.

The disturbing functions will be now transformed to expressions involving only the angle between the location vectors and the ratios between the semi-major axis and distance from the central body. This will allow easy expansion in terms of orbital elements. Let us denote

$$\mathcal{S} := \frac{a'}{\overline{\mu}'}\mathcal{R} = \mathcal{S}_D + \alpha\mathcal{S}_E \tag{17}$$

$$\mathcal{S}' := \frac{a}{\overline{\mu}}\mathcal{R}' = \alpha\mathcal{S}_D + \alpha^{-1}\mathcal{S}_I, \tag{18}$$

where $\alpha := a/a'$ and

$$\mathcal{S}_D = \frac{a'}{|\boldsymbol{r}-\boldsymbol{r}'|} = a'\left(r'^2 - 2r'r\cos\psi + r^2\right)^{-1/2} \tag{19}$$

$$\mathcal{S}_E = -\frac{a'}{\overline{\mu}'}\left(\frac{a'}{a}\right)\frac{rr'\cos\psi}{r'^2} = -\left(\frac{r}{a}\right)\left(\frac{a'}{r'}\right)^2\cos\psi, \tag{20}$$

$$\mathcal{S}_I = -\frac{a}{\overline{\mu}}\left(\frac{a}{a'}\right)\frac{r'r\cos\psi}{r^2} = -\left(\frac{r'}{a'}\right)\left(\frac{a}{r}\right)^2\cos\psi \tag{21}$$

where ψ is the angle between the vectors $\boldsymbol{r}$ and $\boldsymbol{r}'$.

1.2 Expansion of the Disturbing Functions

Let us denote

$$\zeta := \frac{r-a}{a}, \quad \zeta' := \frac{r'-a'}{a'}, \quad \delta := \cos\psi - \cos(\theta-\theta'), \tag{22}$$

where $\theta = \varpi + f, \theta' = \varpi' + f'$ are the true longitudes for the two planets.

The variables ζ and ζ' essentially measure how many multiples of semi-major axis the corresponding body is away from a circular orbit with with semi-major axis a. For nearly circular orbits, ζ, ζ' are very small. In general, for elliptic orbits $r_{max} = a(1+e)$ and $\zeta_{max} = [a(1+e)-a]/a = e$. The variable δ disappears when the planets orbit on the same plane and measures the difference in ψ due to difference in orbital planes. We assume that the inclinations and eccentrities of both planets are small, which suggests that the effect of e and I can be included using a series expansion.

The equations (19)-(21) are first expanded in terms of Keplerian elements $e, \lambda, \Omega, \varpi$ and $s := \sin(I/2)$. The series expansions here are done up to the second order of eccentricity or inclination. That is, we only include terms proportional to $e^{k_1}e'^{k_2}s^{k_3}s'^{k_4}$, where $k_1 + k_2 + k_3 + k_4 \le 2$. To avoid too complicated expressions in equations, we often compute parts of an expression involving specific orders of (e, e, s, s') separately and use a dedicated notation to indicate this. For example, $\cos[j(\theta-\theta')]$ will involve orders $1, e, e', ee', e^2, e'^2$. The part of the expression that is proportional to e^2, will be denoted by $\cos[j(\theta-\theta')]_{e^2}$. That is, in a second order expression

$$\begin{aligned}\cos[j(\theta-\theta')] &= \cos[j(\theta-\theta')]_1 \\ &+ e\cos[j(\theta-\theta')]_e + e'\cos[j(\theta-\theta')]_{e'} \\ &+ ee'\cos[j(\theta-\theta')]_{ee'} + e^2\cos[j(\theta-\theta')]_{e^2} + e'^2\cos[j(\theta-\theta')]_{e'^2}.\end{aligned} \tag{23}$$

1.2.1 Taylor Series Expansion of $\mathcal{S}_D$

Substitution of $a' = r'/(1+\zeta')$ to (19), yields

$$\mathcal{S}_D = \frac{1}{1+\zeta'}\left(\frac{r^2}{r'^2} - 2\frac{r}{r'}\cos\psi + 1\right)^{-1/2}. \tag{24}$$

The first two terms can be expanded

$$\frac{r}{r'} = \frac{a(1+\zeta)}{a'(1+\zeta')} = \alpha\frac{1+\zeta}{1+\zeta'}, \tag{25}$$

$$2\frac{r}{r'}\cos\psi = 2\alpha\frac{1+\zeta}{1+\zeta'}\cos\psi = 2\alpha\frac{1+\zeta}{1+\zeta'}\delta - 2\alpha\frac{1+\zeta}{1+\zeta'}\cos(\theta-\theta') \tag{26}$$

to obtain

$$S_D = \left[\alpha^2 \left(\frac{1+\zeta}{1+\zeta'}\right)^2 - 2\alpha\frac{1+\zeta}{1+\zeta'}\delta - 2\alpha\frac{1+\zeta}{1+\zeta'}\cos(\theta-\theta') + 1\right]^{-1/2}. \tag{27}$$

When the orbits are circular, the expressions $(1+\zeta)/(1+\zeta')$ simplify to unity. Since the planetary orbits are nearly circular and are close to the same plane, it makes expand (27) as a Taylor series w.r.t. perfectly circular orbits on the zero-inclination plane. Define

$$\tilde{\alpha} := \frac{1+\zeta}{1+\zeta'}\alpha. \tag{28}$$

Define

$$F(\alpha, \Delta, \theta-\theta') := \left[1+\alpha^2 - 2\alpha\Delta - 2\alpha\cos(\theta-\theta')\right]^{-1/2}. \tag{29}$$

Then, (27) can be rewritten $S_D = F(\tilde{\alpha}, \delta, \theta-\theta')$ and circular orbits correspond to $F(\alpha, \delta, \theta-\theta')$ and planets on same zero-inclination plane correspond to $F(\tilde{\alpha}, 0)$. We expand to the second order

$$\begin{aligned} F(\tilde{\alpha}, \delta, \theta-\theta') &\approx F(\alpha, 0, \theta-\theta') + \frac{\partial F}{\partial\alpha}(\alpha, 0, \theta-\theta')(\tilde{\alpha}-\alpha) + \frac{1}{2}\frac{\partial^2 F}{\partial\alpha^2}(\alpha, 0, \theta-\theta')(\tilde{\alpha}-\alpha)^2 \\ &+ \frac{\partial F}{\partial\Delta}(\alpha, 0, \theta-\theta')\delta. \end{aligned} \tag{30}$$

The last term can be evaluated

$$\frac{\partial F}{\partial\Delta}\delta = -\frac{1}{2}\frac{-2\alpha}{[\cdot]^3}\delta = \alpha F^3\delta \tag{31}$$

Denoting $D := \partial/\partial\alpha$, we can approximate

$$S_D \approx \frac{1}{1+\zeta'}\left[1 + (\tilde{\alpha}-\alpha)D + \frac{1}{2}(\tilde{\alpha}-\alpha)^2 D^2\right] F + \delta\alpha F^3. \tag{32}$$

We also approximate to the second order

$$\frac{1}{1+\zeta'} = \sum_{k=0}^{\infty}(-1)^k\zeta'^k \approx 1 - \zeta' + \zeta'^2 + \mathcal{O}(\zeta'^3), \tag{33}$$

$$\frac{1}{(1+\zeta')^2} = \sum_{k=0}^{\infty}(-1)^k(1+n)\zeta'^k \approx 1 - 2\zeta' + 3\zeta'^2 + \mathcal{O}(\zeta'^3), \tag{34}$$

$$\frac{1}{(1+\zeta')^3} = \frac{1}{2}\sum_{k=0}^{\infty}(-1)^n(n+1)(n+2)\zeta'^n \approx 1 - 3\zeta' + 6\zeta'^2 + \mathcal{O}(\zeta^3) \tag{35}$$

to obtain

$$\frac{\tilde{\alpha}-\alpha}{1+\zeta'} = \frac{\alpha}{1+\zeta'}\left(\frac{1+\zeta}{1+\zeta'} - 1\right) = \alpha\frac{\zeta-\zeta'}{(1+\zeta')^2} \approx \alpha(\zeta-\zeta')(1-2\zeta') = \alpha(\zeta-\zeta'+2\zeta'^2-2\zeta\zeta'), \tag{36}$$

$$\frac{(\tilde{\alpha}-\alpha)^2}{1+\zeta'} = \frac{\alpha^2}{1+\zeta'}\left(\frac{1+\zeta}{1+\zeta'} - 1\right)^2 = \alpha^2\frac{(\zeta-\zeta')^2}{(1+\zeta')^3} \approx \alpha^2(\zeta^2-2\zeta\zeta'+\zeta'^2)(1-3\zeta') \approx \alpha^2(\zeta^2-2\zeta\zeta'+\zeta'^2). \tag{37}$$

Therefore, the Taylor series expansion (32) can be written

$$S_D \approx \left[(1-\zeta'+\zeta'^2) + (\zeta-\zeta'+2\zeta'^2-2\zeta\zeta')\alpha D + \frac{1}{2}(\zeta^2-2\zeta\zeta'+\zeta'^2)\alpha^2 D^2\right] F + \delta\alpha F^3 \tag{38}$$

1.2.2 Fourier Series Expansion

Since F is a function of the true longitude difference $\theta - \theta'$, it makes sense to expand the angular dependence as a Fourier series. This dependence is symmetric so the expansion will contain only the cosine terms. That is,

$$F(\alpha, 0, \theta - \theta') = \frac{1}{2} \sum_{j=-\infty}^{\infty} b^j_{1/2}(\alpha) \cos[j(\theta - \theta')], \tag{39}$$

$$F^3(\alpha, 0, \theta - \theta') = \frac{1}{2} \sum_{j=-\infty}^{\infty} b^j_{3/2}(\alpha) \cos[j(\theta - \theta')], \tag{40}$$

where **Laplace coefficients**

$$b^{(j)}_s := \frac{1}{\pi} \int_0^{2\pi} \frac{\cos j\psi \, d\psi}{(1 - 2\alpha \cos\psi + \alpha^2)^s}. \tag{41}$$

Thereafter, (38) can be expanded

$$\begin{aligned} \mathcal{S}_D &\approx \frac{1}{2} \sum_{j=-\infty}^{\infty} \left\{ \left[(1 - \zeta' + \zeta'^2) + (\zeta - \zeta' + 2\zeta'^2 - 2\zeta\zeta')\alpha D + \frac{1}{2}(\zeta^2 - 2\zeta\zeta' + \zeta'^2)\alpha^2 D^2 \right] b^{(j)}_{1/2}(\alpha) \right. \\ &+ \left. \delta\alpha b^{(j)}_{3/2}(\alpha) \right\} \cos[j(\theta - \theta')]. \end{aligned} \tag{42}$$

We wish to expand the operator in square brackets as a function $G(\lambda, \lambda', \varpi, \varpi', \alpha, \alpha')$. Thus, employ the approximations

$$\zeta = -e\cos(\lambda - \varpi) + \frac{1}{2}e^2 - \frac{1}{2}e^2\cos(2\lambda - 2\varpi) + \mathcal{O}(e^3), \tag{43}$$

$$\zeta^2 = e^2\cos^2(\lambda - \varpi) + \mathcal{O}(e^3) \tag{44}$$

$$\approx \frac{1}{2}e^2 + \frac{1}{2}e^2\cos(2\lambda - 2\varpi), \tag{45}$$

$$\zeta\zeta' = ee'\cos(\lambda - \varpi)\cos(\lambda' - \varpi') + \mathcal{O}(e^3) \tag{46}$$

$$\approx \frac{1}{2}ee'\cos(\lambda - \lambda' - \varpi + \varpi') + \frac{1}{2}ee'\cos(\lambda + \lambda' - \varpi - \varpi') \tag{47}$$

where we have used the trigonometric identity $\cos 2x = 2\cos^2 x - 1$. Thus, we can expand

$$\begin{aligned} 1 - \zeta' + \zeta'^2 &= 1 + e'\cos(\lambda' - \varpi') - \frac{1}{2}e'^2 + \frac{1}{2}e'^2\cos(2\lambda' - 2\varpi') \\ &+ \frac{1}{2}e'^2 + \frac{1}{2}e'^2\cos(2\lambda' - 2\varpi') \\ &= 1 + e'\cos(\lambda' - \omega') + e'^2\cos(2\lambda' - 2\omega'), \end{aligned} \tag{48, 49}$$

$$\begin{aligned} \zeta - \zeta' + 2\zeta'^2 - 2\zeta\zeta' &= -e\cos(\lambda - \varpi) + \frac{1}{2}e^2 - \frac{1}{2}e^2\cos(2\lambda - 2\varpi) \\ &+ e'\cos(\lambda' - \varpi') - \frac{1}{2}e'^2 + \frac{1}{2}e'^2\cos(2\lambda' - 2\varpi') \\ &+ e'^2 + e'^2\cos(2\lambda' - 2\varpi') \\ &- ee'\cos(\lambda - \lambda' - \varpi + \varpi') - ee'\cos(\lambda + \lambda' - \varpi - \varpi') \end{aligned} \tag{50}$$

$$\begin{aligned}\frac{1}{2}(\zeta^2 - 2\zeta\zeta' + \zeta'^2) &= \frac{1}{4}e^2 + \frac{1}{4}e^2\cos(2\lambda - 2\varpi) \\ &- \frac{1}{2}ee'\cos(\lambda - \lambda' - \varpi + \varpi') - \frac{1}{2}ee'\cos(\lambda + \lambda' - \varpi - \varpi') \\ &+ \frac{1}{4}e'^2 + \frac{1}{4}e'^2\cos(2\lambda' - 2\varpi')\end{aligned} \tag{51}$$

Again, we can expand the components

$$G_1 = 1, \tag{52}$$
$$G_e = -\cos(\lambda - \varpi)\alpha D, \tag{53}$$
$$G_{e'} = \cos(\lambda' - \varpi')(1 + \alpha D), \tag{54}$$
$$G_{ee'} = \left[\cos(\lambda - \lambda' - \varpi + \varpi') + \cos(\lambda + \lambda' - \varpi - \varpi')\right](-\alpha D - \alpha^2 D^2/2), \tag{55}$$
$$G_{e^2} = \frac{1}{2}\left[1 - \cos(2\lambda - 2\varpi)\right]\alpha D + \frac{1}{4}\left[1 + \cos(2\lambda - 2\varpi)\right]\alpha^2 D^2, \tag{56}$$
$$G_{e'^2} = \cos(2\lambda' - 2\omega') + \left[\frac{1}{2} + \frac{3}{2}\cos(2\lambda' - 2\varpi')\right]\alpha D + \frac{1}{4}[1 + \cos(2\lambda' - 2\varpi')]\alpha^2 D^2. \tag{57}$$

Thus, we can expand

$$\begin{aligned}G &= 1 \\ &- e\cos(\lambda - \varpi)\alpha D \\ &+ e'\cos(\lambda' - \varpi')(1 + \alpha D) \\ &+ ee'\left[\cos(\lambda - \lambda' - \varpi + \varpi') + \cos(\lambda + \lambda' - \varpi - \varpi')\right](-\alpha D - \alpha^2 D^2/2) \\ &+ e^2\left\{\frac{1}{2}\left[1 - \cos(2\lambda - 2\varpi)\right]\alpha D + \frac{1}{4}\left[1 + \cos(2\lambda - 2\varpi)\right]\alpha^2 D^2\right\} \\ &+ e'^2\left\{\cos(2\lambda' - 2\omega') + \left[\frac{1}{2} + \frac{3}{2}\cos(2\lambda' - 2\varpi')\right]\alpha D + \frac{1}{4}[1 + \cos(2\lambda' - 2\varpi')]\alpha^2 D^2\right\}.\end{aligned} \tag{58}$$

1.2.3 Expansion of the Inclination Term

We wish to expand the inclination term in (42)

$$\delta = \cos\psi - \cos(\theta - \theta'). \tag{59}$$

To derive an expression for $\cos\psi$, we first expand the position of a planet

$$X/r = \cos\Omega\cos(\omega + f) - \sin\Omega\sin(\omega + f)\cos I, \tag{60}$$
$$Y/r = \sin\Omega\cos(\omega + f) + \cos\Omega\sin(\omega + f)\cos I, \tag{61}$$
$$Z/r = \sin(\omega + f)\sin I. \tag{62}$$

We will write $s := \sin\frac{I}{2}$, where s is considered $\mathcal{O}(e)$. Then,

$$\cos I = 1 - 2\sin^2\frac{I}{2} = 1 - 2s^2 \tag{63}$$
$$\sin I = 2\sin\frac{I}{2}\cos\frac{I}{2} = 2\sin\frac{I}{2}\sqrt{1 - \sin^2\frac{I}{2}} \approx 2s \tag{64}$$

Then,

$$\cos\psi \;=\; \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{rr'} = \frac{XX'+YY'+ZZ'}{rr'}. \tag{65}$$

To compute this expression, we need second order expansions for $\cos(\omega+f)$ and $\sin(\omega+f)$ in terms of M and e. We will use the expansions ($TODO$)

$$\begin{aligned}
\sin f &= \sin M + e\sin 2M + e^2/8(9\sin 3M - 7\sin M) + \mathcal{O}(e^3), &(66)\\
\cos f &= \cos M + e(\cos 2M - 1) + e^2/8(9\cos 3M - 9\cos M) + \mathcal{O}(e^3). &(67)
\end{aligned}$$

Then,

$$\begin{aligned}
\cos(\omega+f) &= \cos\omega\cos f - \sin\omega\sin f &(68)\\
&= \cos\omega[\cos M + e(\cos 2M - 1) + \frac{e^2}{8}(9\cos 3M - 9\cos M)] &(69)\\
&- \sin\omega[\sin M + e\sin 2M + \frac{e^2}{8}(9\sin 3M - 7\sin M)]\\
&= \cos\omega\cos M - \sin\omega\sin M &(70)\\
&+ e(\cos\omega\cos 2M - \sin\omega\sin 2M - \cos\omega)\\
&+ \frac{e^2}{8}(9\cos\omega\cos 3M - 9\sin\omega\sin 3M - 9\cos\omega\cos M + 7\sin\omega\sin M)
\end{aligned}$$

We can substitute

$$9\cos\omega\cos M - 7\sin\omega\sin M = 8(\cos\omega\cos M - 8\sin\omega\sin M) + (\cos\omega\cos M + \sin\omega\sin M) \tag{71}$$

to obtain

$$\begin{aligned}
\cos(\omega+f) &= \cos(\omega+M) &(72)\\
&+ e[\cos(\omega+2M) - \cos\omega]\\
&+ \frac{e^2}{8}\left[9\cos(\omega+3M) - 8\cos(\omega+M) - \cos(\omega-M)\right].
\end{aligned}$$

Similarly

$$\begin{aligned}
\sin(\omega+f) &= \sin\omega\cos f + \cos\omega\sin f &(73)\\
&= \sin\omega[\cos M + e(\cos 2M - 1) + e^2/8(9\cos 3M - 9\cos M)] &(74)\\
&+ \cos\omega[\sin M + e\sin 2M + e^2/8(9\sin 3M - 7\sin M)]\\
&= \sin\omega\cos M + \cos\omega\sin M &(75)\\
&+ e\left(\sin\omega\cos 2M + \cos\omega\sin 2M - \sin\omega\right)\\
&+ \frac{e^2}{8}\left[9\sin\omega\cos 3M + 9\cos\omega\sin 3M - 9\sin\omega\cos M - 7\cos\omega\sin M\right]
\end{aligned}$$

We can substitute

$$9\sin\omega\cos M + 7\cos\omega\sin M = 8(\sin\omega\cos M + \cos\omega\sin M) + \sin\omega\cos M - \cos\omega\sin M \tag{76}$$

to obtain

$$\begin{aligned} \sin(\omega + f) &= \sin(\omega + M) \\ &+ e[\sin(\omega + 2M) - \sin\omega] \\ &+ \frac{e^2}{8}\left[9\sin(\omega + 3M) - 8\sin(\omega + M) - \sin(\omega - M)\right]. \end{aligned} \tag{77}$$

The expressions $X/r, Y/r, Z/r$ now follow

$$
\begin{aligned}
(X/r)_1 &= \cos\Omega\cos(\omega+f)_1 - \sin\Omega\sin(\omega+f)_1(\cos I)_1 && (78)\\
&= \cos\Omega\cos(\omega+M) - \sin\Omega\sin(\omega+M) && (79)\\
&= \cos(\Omega+\omega+M), && (80)\\
(Y/r)_1 &= \sin\Omega\cos(\omega+f)_1 + \cos\Omega\sin(\omega+f)_1(\cos I)_1 && (81)\\
&= \sin\Omega\cos(\omega+M) + \cos\Omega\sin(\omega+M) && (82)\\
&= \sin(\Omega+\omega+M), && (83)\\
(Z/r)_1 &= \sin(\omega+f)_1(\sin I)_1 && (84)\\
&= 0, && (85)\\
(X/r)_s &= 0, && (86)\\
(Y/r)_s &= 0, && (87)\\
(Z/r)_s &= \sin(\omega+f)_1(\sin I)_s && (88)\\
&= 2\sin(\omega+M), && (89)\\
(X/r)_{s^2} &= -\sin\Omega\sin(\omega+f)_1(\cos I)_{s^2} && (90)\\
&= 2\sin\Omega\sin(\omega+M) && (91)\\
&= \cos(\Omega-\omega-M) - \cos(\Omega+\omega+M), && (92)\\
(Y/r)_{s^2} &= \cos\Omega\sin(\omega+f)_1(\cos I)_{s^2} && (93)\\
&= -2\cos\Omega\sin(\omega+M) && (94)\\
&= \sin(\Omega-\omega-M) - \sin(\Omega+\omega+M), && (95)\\
(Z/r)_{s^2} &= 0, && (96)\\
(X/r)_e &= \cos\Omega\cos(\omega+f)_e - \sin\Omega\sin(\omega+f)_e(\cos I)_1 && (97)\\
&= \cos\Omega[\cos(\omega+2M) - \cos\omega] - \sin\Omega[\sin(\omega+2M) - \sin\omega] && (98)\\
&= \cos(\Omega+\omega+2M) - \cos(\Omega+\omega) && (99)\\
(Y/r)_e &= \sin\Omega\cos(\omega+f)_e + \cos\Omega\sin(\omega+f)_e(\cos I)_1 && (100)\\
&= \sin\Omega[\cos(\omega+2M) - \cos\omega] + \cos\Omega[\sin(\omega+2M) - \sin\omega] && (101)\\
&= \sin(\Omega+\omega+2M) - \sin(\Omega+\omega) && (102)\\
(Z/r)_e &= \sin(\omega+f)_e\sin(I)_1 && (103)\\
&= 0, && (104)\\
(X/r)_{es} &= \cos\Omega\cos(\omega+f)_e - \sin\Omega\sin(\omega+f)_e(\cos I)_s && (105)\\
&= 0, && (106)\\
(Y/r)_{es} &= \sin\Omega\cos(\omega+f)_e + \cos\Omega\sin(\omega+f)_e(\cos I)_s && (107)\\
&= 0, && (108)\\
(Z/r)_{es} &= \sin(\omega+f)_e(\sin I)_s && (109)\\
&= 2\cos(\omega+2M) - 2\cos\omega && (110)
\end{aligned}
$$

$$
\begin{aligned}
(X/r)_{e^2} &= \cos\Omega\cos(\omega+f)_{e^2} - \sin\Omega\sin(\omega+f)_{e^2}(\cos I)_1 && (111)\\
&= \frac{1}{8}\cos\Omega\,[9\cos(\omega+3M) - 8\cos(\omega+M) - \cos(\omega-M)] && (112)\\
&- \frac{1}{8}\sin\Omega\,[9\sin(\omega+3M) - 8\sin(\omega+M) - \sin(\omega-M)] \\
&= \frac{9}{8}\cos(\Omega+\omega+3M) - \cos(\Omega+\omega+M) - \frac{1}{8}\cos(\Omega+\omega-M) && (113)\\
(Y/r)_{e^2} &= \sin\Omega\cos(\omega+f)_{e^2} + \cos\Omega\sin(\omega+f)_{e^2}(\cos I)_1 && (114)\\
&= \frac{1}{8}\sin\Omega\,[9\cos(\omega+3M) - 8\cos(\omega+M) - \cos(\omega-M)] && (115)\\
&+ \frac{1}{8}\cos\Omega\,[9\sin(\omega+3M) - 8\sin(\omega+M) - \sin(\omega-M)] \\
&= \frac{9}{8}\sin(\Omega+\omega+3M) - \sin(\Omega+\omega+M) - \frac{1}{8}\sin(\Omega+\omega-M), && (116)\\
(Z/r)_{e^2} &= \sin(\omega+f)_{e^2}(\sin I)_1 && (117)\\
&= 0. && (118)
\end{aligned}
$$

Now

$$
\begin{aligned}
(\cos\psi)_1 &= (X/r)_1(X'/r')_1 + (Y/r)_1(Y'/r')_1 + (Z/r)_1(Z'/r')_1 && (119)\\
&= \cos(\Omega+\omega+M)\cos(\Omega'+\omega'+M') + \sin(\Omega+\omega+M)\sin(\Omega'+\omega'+M') && (120)\\
&= \cos(\Omega+\omega+M-\Omega'-\omega'-M') && (121)\\
&= \cos(\lambda-\lambda') && (122)\\
(\cos\psi)_s &= (Z/r)_s(Z'/r')_1 = 0 && (123)\\
(\cos\psi)_{s'} &= (Z/r)_1(Z'/r')_{s'} = 0 && (124)\\
(\cos\psi)_{ss'} &= (Z/r)_s(Z'/r')_{s'} && (125)\\
&= 4\sin(\omega+M)\sin(\omega'+M') && (126)\\
&= 2\cos(\lambda-\lambda'-\Omega+\Omega') - 2\cos(\lambda+\lambda'-\Omega-\Omega') && (127)\\
(\cos\psi)_{s^2} &= (X/r)_{s^2}(X'/r')_1 + (Y/r)_{s^2}(Y'/r')_1 && (128)\\
&= [\cos(\Omega-\omega-M) - \cos(\Omega+\omega+M)]\cos(\Omega'+\omega'+M') && (129)\\
&+ [\sin(\Omega-\omega-M) - \sin(\Omega+\omega+M)]\sin(\Omega'+\omega'+M') \\
&= \cos(\Omega-\omega-M-\Omega'-\omega'-M) - \cos(\Omega+\omega+M-\Omega'-\omega'-M) && (130)\\
&= \cos(\lambda+\lambda'-2\Omega) - \cos(\lambda-\lambda') && (131)\\
(\cos\psi)_{s'^2} &= \cos(\lambda+\lambda'-\Omega') - \cos(\lambda-\lambda') && (132)\\
(\cos\psi)_e &= (X/r)_e(X'/r')_1 + (Y/r)_e(Y'/r')_1 && (133)\\
&= [\cos(\lambda+M) - \cos(\lambda-M)]\cos\lambda' + [\sin(\lambda+M) - \sin(\lambda-M)]\sin\lambda' && (134)\\
&= \cos(\lambda-\lambda'+M) - \cos(\lambda-\lambda'-M) && (135)\\
&= \cos(2\lambda-\lambda'-\varpi) - \cos(\lambda'-\varpi), && (136)\\
(\cos\psi)_{e'} &= \cos(2\lambda'-\lambda-\varpi') - \cos(\lambda-\varpi') && (137)\\
&= \cos(\lambda-2\lambda'+\varpi') - \cos(\lambda-\varpi'), && (138)
\end{aligned}
$$

$$
\begin{aligned}
(\cos\psi)_{es} &= (Z/r)_{es}(Z'/r')_1 = 0, && (139)\\
(\cos\psi)_{e's'} &= (Z/r)_1(Z'/r')_{e's'} = 0, && (140)\\
(\cos\psi)_{ee'} &= (X/r)_e(X'/r')_{e'} + (Y/r)_e(Y'/r')_{e'} && (141)\\
&= [\cos(\lambda+M)-\cos(\lambda-M)]\left[\cos(\lambda'+M')-\cos(\lambda'-M')\right] && (142)\\
&+ [\sin(\lambda+M)-\sin(\lambda-M)]\left[\sin(\lambda'+M')-\sin(\lambda'-M')\right]\\
&= \cos(\lambda+M-\lambda'-M')-\cos(\lambda+M-\lambda'+M') && (143)\\
&- \cos(\lambda-M-\lambda'-M')+\cos(\lambda-M-\lambda'+M')\\
&= \cos(2\lambda-2\lambda'-\varpi+\varpi')-\cos(2\lambda-\varpi-\varpi') && (144)\\
&- \cos(2\lambda'-\varpi'-\varpi)+\cos(\varpi-\varpi')\\
(\cos\psi)_{e^2} &= (X/r)_{e^2}(X'/r')_1 + (Y/r)_{e^2}(Y'/r')_1 && (145)\\
&= \left[\frac{9}{8}\cos(\Omega+\omega+3M)-\cos(\Omega+\omega+M)-\frac{1}{8}\cos(\Omega+\omega-M)\right]\cos\lambda' && (146)\\
&+ \left[\frac{9}{8}\sin(\Omega+\omega+3M)-\sin(\Omega+\omega+M)-\frac{1}{8}\sin(\Omega+\omega-M)\right]\sin\lambda'\\
&= \frac{9}{8}\cos(\lambda+2M-\lambda')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda-2M-\lambda') && (147)\\
&= \frac{9}{8}\cos(3\lambda-\lambda'-2\varpi)-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi), && (148)\\
(\cos\psi)_{e'^2} &= \frac{9}{8}\cos(3\lambda'-\lambda-2\varpi')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda'+\lambda-2\varpi'), && (149)
\end{aligned}
$$

Collecting the terms yields

$$
\begin{aligned}
\cos\psi &= \cos(\lambda-\lambda') && (150)\\
&+ e[\cos(2\lambda-\lambda'-\varpi)-\cos(\lambda'-\varpi)]\\
&+ e'[\cos(2\lambda'-\lambda-\varpi')-\cos(\lambda-\varpi')]\\
&+ ee'[\cos(2\lambda-2\lambda'-\varpi+\varpi')-\cos(2\lambda-\varpi-\varpi')-\cos(2\lambda'-\varpi'-\varpi)+\cos(\varpi-\varpi')]\\
&+ e^2\left[\frac{9}{8}\cos(3\lambda-\lambda'-2\varpi)-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi)\right]\\
&+ e'^2\left[\frac{9}{8}\cos(3\lambda'-\lambda-2\varpi')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda'+\lambda-2\varpi')\right]\\
&+ 2ss\left[2\cos(\lambda-\lambda'-\Omega+\Omega')-2\cos(\lambda+\lambda'-\Omega-\Omega')\right]\\
&+ s^2\left[\cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda')\right]\\
&+ s'^2\left[\cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda')\right].
\end{aligned}
$$

When inclination for both planets is equal to 0, we get $s = s' = 0$ and $\cos\psi = \cos(\theta-\theta')$. Moreover inclination doesn't influence the true anomaly. Thus,

$$
\begin{aligned}
\delta &= 2ss'\left[\cos(\lambda-\lambda'-\Omega+\Omega')-\cos(\lambda+\lambda'-\Omega-\Omega')\right] && (151)\\
&+ s^2\left[\cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda')\right]\\
&+ s'^2\left[\cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda')\right].
\end{aligned}
$$

1.2.4 Expansion of $\cos[j(\theta - \theta')]$

We wish to express $\cos[j(\theta-\theta')]$ as a function of the mean longitudes $\lambda = \Omega+\omega+M$, $\lambda' = \Omega'+\omega'+M'$ and the longitudes of perihelion $\varpi = \Omega+\omega$, $\varpi' = \Omega'+\omega'$. At small eccentricities the difference between the true anomaly f and the mean anomaly M is small. Thus, we use the second order series expansion

$$f = M + 2e\sin M + \frac{5}{4}e^2 \sin 2M + \mathcal{O}(e^3) \tag{152}$$

and second order Taylor series expansion

$$\cos x = \cos x_0 - \sin x_0(x-x_0) - \frac{1}{2}\cos x_0 (x-x_0)^2 + \mathcal{O}[(x-x_0)^3], \tag{153}$$

$$\sin x = \sin x_0 + \cos x_0(x-x_0) - \frac{1}{2}\sin x_0 (x-x_0)^2 + \mathcal{O}[(x-x_0)^3], \tag{154}$$

where $x = jf$ and $x_0 = jM$. The computation is carried out as follows:

1. Second-order expansions for $\sin jf$ and $\cos jf$.
2. $\cos j\theta$ and $\sin j\theta$ using trigonometric addition formulas with ϖ.
3. $\cos[j(\theta-\theta')]$ with trigonometric subtraction formula.

That, we will start by expanding:

$$\cos jf \approx \cos jM - \sin jM[j(f-M)] - \frac{1}{2}\cos jM[j(f-M)]^2 \tag{155}$$

$$= \cos jM - \sin jM\left[j\left(2e\sin M + \frac{5}{4}e^2\sin 2M\right)\right] - \frac{1}{2}\cos jM\left[j^2(4e^2\sin^2 M)\right] \tag{156}$$

$$= \cos jM - 2ej\sin jM\sin M - e^2\left[\frac{5}{4}j\sin jM\sin 2M + j^2\cos jM(1-\cos 2M)\right], \tag{157}$$

$$\sin jf \approx \sin jM + \cos jM(f-M) - \frac{1}{2}\sin jM(f-M)^2 \tag{158}$$

$$= \sin jM + \cos jM\left[j\left(2e\sin M + \frac{5}{4}e^2\sin 2M\right)\right] - \frac{1}{2}\sin jM\left[j^2\left(4e^2\sin^2 M\right)\right] \tag{159}$$

$$= \sin jM + 2ej\cos jM\sin M + e^2\left[\frac{5}{4}j\cos jM\sin 2M - j^2\sin jM\,(1-\cos 2M)\right] \tag{160}$$

where we have used the trigonometric formula $2\sin^2 M = 1-\cos 2M$. That is, specific orders can be written

$$(\cos jf)_1 = \cos jM, \tag{161}$$

$$(\cos jf)_e = -2j\sin jM\sin M, \tag{162}$$

$$(\cos jf)_{e^2} = -\frac{5}{4}j\sin jM\sin 2M - j^2\cos jM(1-\cos 2M), \tag{163}$$

$$(\sin jf)_1 = \sin jM, \tag{164}$$

$$(\sin jf)_e = 2j\cos jM\sin M, \tag{165}$$

$$(\sin jf)_{e^2}, = \frac{5}{4}j\cos jM\sin 2M - j^2\sin jM\,(1-\cos 2M)\,. \tag{166}$$

Let us now compute the expressions for the true anomaly

$$\cos j\theta = \cos j\varpi \cos jf - \sin j\varpi \sin jf \tag{167}$$
$$\sin j\theta = \sin j\varpi \cos jf + \cos j\varpi \sin jf. \tag{168}$$

This is most conveninent to do separately for the orders $1, e, e^2$ as follows:

$$\begin{aligned}
(\cos j\theta)_1 &= \cos j\varpi (\cos jf)_1 - \sin j\varpi (\sin jf)_1 && (169)\\
&= \cos j\varpi \cos jM - \sin j\varpi \sin jM && (170)\\
&= \cos[j(\varpi + M)] && (171)\\
&= \cos j\lambda, && (172)\\
(\sin j\theta)_1 &= \sin j\varpi (\cos jf)_1 + \cos j\varpi (\sin jf)_1 && (173)\\
&= \sin j\varpi \cos jM + \cos j\varpi \sin jM && (174)\\
&= \sin[j(\varpi + f)] && (175)\\
&= \sin j\lambda. && (176)
\end{aligned}$$

$$\begin{aligned}
(\cos j\theta)_e &= \cos j\varpi (\cos jf)_e - \sin j\varpi (\sin jf)_e && (177)\\
&= \cos j\varpi \,(-2 \sin jM \sin M) - \sin j\varpi \,(2j \cos jM \sin M) && (178)\\
&= -2j \sin M \,(\cos j\varpi \sin jM + \sin j\varpi \cos jM) && (179)\\
&= -2j \sin M \sin j\lambda && (180)\\
&= j \cos(j\lambda + M) - j \cos(j\lambda - M) && (181)\\
&= j \cos[(1+j)\lambda - \varpi] - j \cos[(1-j)\lambda + \varpi], && (182)\\
(\sin j\theta)_e &= \sin j\varpi (\cos jf)_e + \cos j\varpi (\sin jf)_e && (183)\\
&= \sin j\varpi (-2j \sin jM \sin M) + \cos j\varpi (2j \cos jM \sin M) && (184)\\
&= 2j \sin M(-\sin j\varpi \sin jM + \cos j\varpi \cos jM) && (185)\\
&= 2j \sin M \cos j\lambda && (186)\\
&= j \sin(M + j\lambda) + j \sin(M - j\lambda) && (187)\\
&= j \sin[(1+j)\lambda - \varpi] + j \sin[(1-j)\lambda - \varpi]. && (188)
\end{aligned}$$

where we have used the trigonometric identities $2 \sin x \sin y = \cos(x-y) - \cos(x+y)$ and $2 \sin x \cos y = \sin(x+y) + \sin(x-y)$.

$$\begin{aligned}
(\cos j\theta)_{e^2} &= \cos j\varpi (\cos jf)_{e^2} - \sin j\varpi (\sin jf)_{e^2} && (189)\\
&= \cos j\varpi \left[-\frac{5}{4} j \sin jM \sin 2M - j^2 \cos jM (1 - \cos 2M)\right] && (190)\\
&- \sin j\varpi \left[\frac{5}{4} j \cos jM \sin 2M - j^2 \sin jM \,(1 - \cos 2M)\right] \\
&= -\frac{5}{4} j \sin 2M \,[\cos j\varpi \sin jM + \sin j\varpi \cos jM] && (191)\\
&- j^2 \,(1 - \cos 2M)\,(\cos j\varpi \cos jM - \sin j\varpi \sin jM) \\
&= -\frac{5}{4} j \sin 2M \sin j\lambda - j^2 (1 - \cos 2M) \cos j\lambda && (192)\\
&= -j^2 \cos j\lambda - \frac{5}{4} j \sin 2M \sin j\lambda + j^2 \cos 2M \cos j\lambda && (193)
\end{aligned}$$

Substitution of

$$\begin{aligned}
\sin 2M \sin j\lambda &= \frac{1}{2}\cos(j\lambda - 2M) - \frac{1}{2}\cos(j\lambda + 2M) && (194)\\
&= \frac{1}{2}\cos[(2-j)\lambda - 2\varpi] - \frac{1}{2}\cos[(2+j)\lambda - 2\varpi], && (195)\\
\cos 2M \cos j\lambda &= \frac{1}{2}\cos(j\lambda - 2M) + \frac{1}{2}\cos(j\lambda + 2M) && (196)\\
&= \frac{1}{2}\cos[(2-j)\lambda - 2\varpi] + \frac{1}{2}\cos[(2+j)\lambda - 2\varpi]. && (197)
\end{aligned}$$

yields

$$(\cos j\theta)_{e^2} = -j^2\cos j\lambda + \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda - 2\varpi] + \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - 2\varpi]. \quad (198)$$

Similarly

$$\begin{aligned}
(\sin j\theta)_{e^2} &= \sin j\varpi(\cos jf)_{e^2} + \cos j\varpi(\sin jf)_{e^2} && (199)\\
&= \sin j\varpi\left[-\frac{5}{4}j\sin jM\sin 2M - j^2\cos jM(1-\cos 2M)\right] && (200)\\
&+ \cos j\varpi\left[\frac{5}{4}j\cos jM\sin 2M - j^2\sin jM\,(1-\cos 2M)\right] && (201)\\
&= -\frac{5}{4}j\sin 2M\,(\sin j\varpi\sin jM - \cos j\varpi\cos jM) && (202)\\
&- j^2(1-\cos 2M)\,(\sin j\varpi\cos jM + \cos j\varpi\sin jM)\\
&= \frac{5}{4}j\sin 2M\cos j\lambda - j^2(1-\cos 2M)\sin j\lambda && (203)
\end{aligned}$$

Substitution of

$$\begin{aligned}
\sin 2M \cos j\lambda &= \frac{1}{2}\sin(2M + j\lambda) + \frac{1}{2}\sin(2M - j\lambda) && (204)\\
&= \frac{1}{2}\sin[(2+j)\lambda - 2\varpi] + \frac{1}{2}\sin[(2-j)\lambda - 2\varpi], && (205)\\
\cos 2M \sin j\lambda &= \frac{1}{2}\sin(2M + j\lambda) - \frac{1}{2}\sin(2M - j\lambda) && (206)\\
&= \frac{1}{2}\sin[(2+j)\lambda - 2\varpi] - \frac{1}{2}\sin[(2-j)\lambda - 2\varpi]. && (207)
\end{aligned}$$

yields

$$(\sin j\theta)_{e^2} = -j^2\sin j\lambda + \left(\frac{5}{8}j - \frac{1}{2}j^2\right)\sin[(2-j)\lambda - 2\varpi] + \left(\frac{5}{8}j + \frac{1}{2}j^2\right)\sin[(2+j)\lambda - 2\varpi]. \quad (208)$$

That is, collecting the results

$$\begin{aligned}
(\cos j\theta)_1 &= \cos j\lambda, && (209)\\
(\sin j\theta)_1 &= \sin j\lambda, && (210)\\
(\cos j\theta)_e &= j\cos[(1+j)\lambda - \varpi] - j\cos[(1-j)\lambda - \varpi], && (211)\\
(\sin j\theta)_e &= j\sin[(1+j)\lambda - \varpi] + j\sin[(1-j)\lambda - \varpi], && (212)\\
(\cos j\theta)_{e^2} &= -j^2\cos j\lambda + \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda - 2\varpi] + \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - 2\varpi]. && (213)\\
(\sin j\theta)_{e^2} &= -j^2\sin j\lambda - \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\sin[(2-j)\lambda - 2\varpi] + \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\sin[(2+j)\lambda - 2\varpi] && (214)
\end{aligned}$$

allows us to expand

$$\begin{aligned}
\cos j\theta &= (1-e^2j^2)\cos j\lambda && (215)\\
&+ ej\cos[(1+j)\lambda-\varpi]-ej\cos[(1-j)\lambda-\varpi],\\
&+ e^2\left(\frac{1}{2}j^2-\frac{5}{8}j\right)\cos[(2-j)\lambda-2\varpi]+e^2\left(\frac{1}{2}j^2+\frac{5}{8}j\right)\cos[(2+j)\lambda-2\varpi].\\
\sin j\theta &= (1-e^2j^2)\sin j\lambda && (216)\\
&+ ej\sin[(1+j)\lambda-\varpi]+ej\sin[(1-j)\lambda-\varpi]\\
&- e^2\left(\frac{1}{2}j^2-\frac{5}{8}j\right)\sin[(2-j)\lambda-2\varpi]+e^2\left(\frac{1}{2}j^2+\frac{5}{8}j\right)\sin[(2+j)\lambda-2\varpi],
\end{aligned}$$

Now the desired result can be finally computed

$$\begin{aligned}
\cos[j(\theta-\theta')] &= \cos j\theta\cos j\theta'+\sin j\theta\sin j\theta' && (217)\\
&= (\cos j\theta)_1(\cos j\theta')_1+(\sin j\theta)_1(\sin j\theta')_1 && (218)\\
&+ e(\cos j\theta)_e(\cos j\theta')_1+e(\sin j\theta)_e(\sin j\theta')_1\\
&+ e'(\cos j\theta)_1(\cos j\theta')_{e'}+e'(\sin j\theta)_1(\sin j\theta')_{e'}\\
&+ ee'(\cos j\theta)_e(\cos j\theta')_{e'}+ee'(\sin j\theta)_e(\sin j\theta')_{e'}\\
&+ e^2(\cos j\theta)_{e^2}(\cos j\theta')_1+e^2(\sin j\theta)_{e^2}(\sin j\theta')_1\\
&+ e'^2(\cos j\theta)_1(\cos j\theta')_{e'^2}+e'^2(\sin j\theta)_1(\sin j\theta')_{e'^2},
\end{aligned}$$

where

$$\begin{aligned}
\cos[j(\theta-\theta')]_1 &= (\cos j\theta)_1(\cos j\theta')_1+(\sin j\theta)_1(\sin j\theta')_1 && (219)\\
&= \cos j\lambda\cos j\lambda'+\sin j\lambda\sin j\lambda' && (220)\\
&= \cos[j(\lambda-\lambda')], && (221)
\end{aligned}$$

$$\begin{aligned}
\cos[j(\theta-\theta')]_e &= (\cos j\theta)_e(\cos j\theta')_1+(\sin j\theta)_e(\sin j\theta')_1 && (222)\\
&= j\cos[(1+j)\lambda-\varpi]\cos j\lambda'-j\cos[(1-j)\lambda-\varpi]\cos j\lambda' && (223)\\
&+ j\sin[(1+j)\lambda-\varpi]\sin j\lambda'+j\sin[(1-j)\lambda-\varpi]\sin j\lambda'\\
&= j\cos[(1+j)\lambda-j\lambda'-\varpi]-j\cos[(1-j)\lambda+j\lambda'-\varpi], && (224)
\end{aligned}$$

$$\begin{aligned}
\cos[j(\theta-\theta')]_{e'} &= (\cos j\theta)_1(\cos j\theta')_{e'}+(\sin j\theta)_1(\sin j\theta')_{e'} && (225)\\
&= j\cos j\lambda\cos[(1+j)\lambda'-\varpi']-j\cos j\lambda\cos[(1-j)\lambda'-\varpi'] && (226)\\
&+ j\sin j\lambda\sin[(1+j)\lambda'-\varpi']+j\sin j\lambda\sin[(1-j)\lambda'-\varpi'],\\
&= j\cos[(1+j)\lambda'-j\lambda-\varpi']-j\cos[(1-j)\lambda'+j\lambda-\varpi'], && (227)
\end{aligned}$$

$$\cos[j(\theta-\theta')]_{ee'} \tag{228}$$

$$= (\cos j\theta)_e(\cos j\theta')_{e'} + (\sin j\theta)_e(\sin j\theta')_{e'}$$

$$= j^2\left\{\cos[(1+j)\lambda-\varpi]-\cos[(1-j)\lambda-\varpi]\right\}\left\{\cos[(1+j)\lambda'-\varpi']-\cos[(1-j)\lambda'-\varpi']\right\} \tag{229}$$

$$+ j^2\left\{\sin[(1+j)\lambda-\varpi]+\sin[(1-j)\lambda-\varpi]\right\}\left\{\sin[(1+j)\lambda'-\varpi']+\sin[(1-j)\lambda'-\varpi']\right\}$$

$$= j^2\cos[(1+j)\lambda-\varpi]\cos[(1+j)\lambda'-\varpi'] - j^2\cos[(1+j)\lambda-\varpi]\cos[(1-j)\lambda'-\varpi'] \tag{230}$$

$$- j^2\cos[(1-j)\lambda-\varpi]\cos[(1+j)\lambda'-\varpi'] + j^2\cos[(1-j)\lambda-\varpi]\cos[(1-j)\lambda'-\varpi']$$

$$+ j^2\sin[(1+j)\lambda-\varpi]\sin[(1+j)\lambda'-\varpi'] + j^2\sin[(1+j)\lambda-\varpi]\sin[(1-j)\lambda'-\varpi']$$

$$+ j^2\sin[(1-j)\lambda-\varpi]\sin[(1+j)\lambda'-\varpi'] + j^2\sin[(1-j)\lambda-\varpi]\sin[(1-j)\lambda'-\varpi']$$

$$= j^2\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] + j^2\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \tag{231}$$

$$- j^2\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] - j^2\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']$$

$$\cos[j(\theta-\theta')]_{e^2} \tag{232}$$

$$= (\cos j\theta)_{e^2}(\cos j\theta')_1 + (\sin j\theta)_{e^2}(\sin j\theta')_1$$

$$= \left[-j^2\cos j\lambda + \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\cos[(2-j)\lambda-2\varpi] + \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\cos[(2+j)\lambda-2\varpi]\right]\cos j\lambda' \tag{233}$$

$$+ \left[-j^2\sin j\lambda - \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\sin[(2-j)\lambda-2\varpi] + \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\sin[(2+j)\lambda-2\varpi]\right]\sin j\lambda'$$

$$= -j^2\left(\cos j\lambda\cos j\lambda' + \sin j\lambda\sin j\lambda'\right) \tag{234}$$

$$+ \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\left\{\cos[(2-j)\lambda-2\varpi]\cos j\lambda' - \sin[(2-j)\lambda-2\varpi]\sin j\lambda'\right\}$$

$$+ \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\left\{\cos[(2+j)\lambda-2\varpi]\cos j\lambda' + \sin[(2+j)\lambda-2\varpi]\sin j\lambda'\right\}$$

$$= -j^2\cos[j(\lambda-\lambda')] \tag{235}$$

$$+ \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\cos[(2-j)\lambda+j\lambda'-2\varpi] + \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\cos[(2+j)\lambda-j\lambda'-2\varpi]$$

The e'^2 case follows by symmetry. Thus, we can again collect the terms

$$\begin{aligned}
\cos[j(\theta-\theta')]_1 &= \cos[j(\lambda-\lambda')], && (236)\\
\cos[j(\theta-\theta')]_e &= j\cos[(1+j)\lambda - j\lambda' - \varpi] - j\cos[(1-j)\lambda + j\lambda' - \varpi], && (237)\\
\cos[j(\theta-\theta')]_{e'} &= j\cos[(1+j)\lambda' - j\lambda - \varpi'] - j\cos[(1-j)\lambda' + j\lambda - \varpi'], && (238)\\
\cos[j(\theta-\theta')]_{ee'} &= j^2\cos[(1+j)\lambda - (1+j)\lambda' - \varpi + \varpi'] && (239)\\
&+ j^2\cos[(1-j)\lambda - (1-j)\lambda' - \varpi + \varpi']\\
&- j^2\cos[(1+j)\lambda + (1-j)\lambda' - \varpi - \varpi']\\
&- j^2\cos[(1-j)\lambda + (1+j)\lambda' - \varpi - \varpi']\\
\cos[j(\theta-\theta')]_{e^2} &= -j^2\cos[j(\lambda-\lambda')] && (240)\\
&+ \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]\\
&+ \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]\\
\cos[j(\theta-\theta')]_{e'^2} &= -j^2\cos[j(\lambda-\lambda')] && (241)\\
&+ \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

to finally expand

$$\begin{aligned}
\cos[j(\theta-\theta')] &= (1 - e^2j^2 - e'^2j^2)\cos[j(\lambda-\lambda')] && (242)\\
&+ ej\cos[(1+j)\lambda - j\lambda' - \varpi] - ej\cos[(1-j)\lambda + j\lambda' - \varpi]\\
&+ e'j\cos[(1+j)\lambda' - j\lambda - \varpi'] - e'j\cos[(1-j)\lambda' + j\lambda - \varpi']\\
&+ ee'j^2\cos[(1+j)\lambda - (1+j)\lambda' - \varpi + \varpi']\\
&+ ee'j^2\cos[(1-j)\lambda - (1-j)\lambda' - \varpi + \varpi']\\
&- ee'j^2\cos[(1+j)\lambda + (1-j)\lambda' - \varpi - \varpi']\\
&- ee'j^2\cos[(1-j)\lambda + (1+j)\lambda' - \varpi - \varpi']\\
&+ e^2\left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]\\
&+ e^2\left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]\\
&+ e'^2\left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ e'^2\left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

1.2.5 Further Expansion of S_D

Now we must compute the product of (58) and (242). This is rather formidable task. However, since most of the terms are of second order and higher-order terms are discarded, the number of product terms is significantly reduced. We have both $\cos[j(\theta-\theta')]_1 = \cos[j(\lambda-\lambda')]$ and $G_1 = 1$. Thus, multiplications

involving a second-order term are trivial. For order 1, the computation is trivial:

$$\left\{G\cos[j(\theta-\theta')]\right\}_1 \quad = \quad \cos[j(\lambda-\lambda')]. \tag{243}$$

Order e:

$$\left\{G\cos[j(\theta-\theta')]\right\}_e = G_e\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_e \tag{244}$$

$$\begin{aligned} G_e\cos[j(\theta-\theta')]_1 &= -\cos(\lambda-\varpi)\alpha D\cos[j(\lambda-\lambda')] && (245)\\ &= \frac{1}{8}(-4\alpha D)\cos((1+j)\lambda - j\lambda' - \varpi) && (246)\\ &+ \frac{1}{8}(-4\alpha D)\cos((1-j)\lambda + j\lambda' - \varpi) \end{aligned}$$

$$\begin{aligned} G_1\cos[j(\theta-\theta')]_e &= j\cos[(1+j)\lambda - j\lambda' - \varpi] - j\cos[(1-j)\lambda + j\lambda' - \varpi] && (247)\\ &= \frac{1}{8}(8j)\cos[(1+j)\lambda - j\lambda' - \varpi] && (248)\\ &+ \frac{1}{8}(-8j)\cos[(1-j)\lambda + j\lambda' - \varpi] \end{aligned}$$

$$\begin{aligned} \therefore \left\{G\cos[j(\theta-\theta')]\right\}_e &= \frac{1}{8}(8j-4\alpha D)\cos((1+j)\lambda - j\lambda' - \varpi) && (249)\\ &+ \frac{1}{8}(-8j-4\alpha D)\cos((1-j)\lambda + j\lambda' - \varpi) \end{aligned}$$

Order e':

$$\left\{G\cos[j(\theta-\theta')]\right\}_{e'} = G_{e'}\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_{e'} \tag{250}$$

$$\begin{aligned} G_{e'}\cos[j(\theta-\theta')]_1 &= \cos(\lambda'-\varpi')(1+\alpha D)\cos[j(\lambda-\lambda')] && (251)\\ &= \frac{1}{8}(4+4\alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi'] && (252)\\ &+ \frac{1}{8}(4+4\alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi'] \end{aligned}$$

$$\begin{aligned} G_1\cos[j(\theta-\theta')]_{e'} &= j\cos[(1+j)\lambda' - j\lambda - \varpi'] - j\cos[(1-j)\lambda' + j\lambda - \varpi'] && (253)\\ &= \frac{1}{8}(8j)\cos[(1+j)\lambda' - j\lambda - \varpi'] && (254)\\ &- \frac{1}{8}(8j)\cos[(1-j)\lambda' + j\lambda - \varpi']. \end{aligned}$$

$$\begin{aligned} \therefore \left\{G\cos[j(\theta-\theta')]\right\}_{e'} &= \frac{1}{8}(4+8j+4\alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi'] && (255)\\ &+ \frac{1}{8}(4-8j+4\alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi'] \end{aligned}$$

Order ee':

$$\begin{aligned}\left\{G\cos[j(\theta-\theta')]\right\}_{ee'} &= G_e\cos[j(\theta-\theta')]_{e'} + G_{e'}\cos[j(\theta-\theta')]_e. \\ &+ G_{e'e'}\cos[j(\theta-\theta')]_1. + G_1\cos[j(\theta-\theta')]_{ee'}.\end{aligned} \tag{256}$$

$$G_e\cos[j(\theta-\theta')]_{e'} \tag{257}$$

$$\begin{aligned}&= -\cos(\lambda-\varpi)\alpha D\left\{j\cos[(1+j)\lambda'-j\lambda-\varpi'] - j\cos[(1-j)\lambda'+j\lambda-\varpi']\right\} \\ &= (1/8)(-4j\alpha D)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'] \qquad (258)\\ &+ (1/8)(-4j\alpha D)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(4j\alpha D)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)(4j\alpha D)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi']\end{aligned}$$

$$G_{e'}\cos[j(\theta-\theta')]_e \tag{259}$$

$$\begin{aligned}&= \cos(\lambda'-\varpi')(1+\alpha D)\left\{j\cos[(1+j)\lambda-j\lambda'-\varpi] - j\cos[(1-j)\lambda+j\lambda'-\varpi]\right\} \\ &= (1/8)(4j+4j\alpha D)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \qquad (260)\\ &+ (1/8)(4j+4j\alpha D)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(-4j-4j\alpha D)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)(-4j-4j\alpha D)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'].\end{aligned}$$

$$G_{e'e'}\cos[j(\theta-\theta')]_1 \tag{261}$$

$$\begin{aligned}&= \left[\cos(\lambda-\lambda'-\varpi+\varpi') + \cos(\lambda+\lambda'-\varpi-\varpi')\right](-\alpha D-\alpha^2D^2/2)\cos[j(\lambda-\lambda')] \\ &= (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \qquad (262)\\ &+ (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']\end{aligned}$$

$$\begin{aligned}G_1\cos[j(\theta-\theta')]_{ee'} &= (1/8)(8j^2)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(8j^2)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(-8j^2)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)(-8j^2)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']\end{aligned} \tag{263}$$

$$\begin{aligned}\therefore \left\{G\cos[j(\theta-\theta')]\right\}_{ee'} &= \frac{1}{8}\left(4j+8j^2-4\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ \frac{1}{8}\left(-4j+8j^2-4\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ \frac{1}{8}\left(4j-8j^2-4\alpha D+8j\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ \frac{1}{8}\left(-4j-8j^2-4\alpha D-8j\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'].\end{aligned} \tag{264}$$

Order e^2:

$$\left\{G\cos[j(\theta-\theta')]\right\}_{e^2} = G_{e^2}\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_{e^2} + G_e\cos[j(\theta-\theta')]_e. \tag{265}$$

$$\begin{aligned}
G_{e^2}\cos[j(\theta-\theta')]_1 &= \left\{\frac{1}{2}\left[1-\cos(2\lambda-2\varpi)\right]\alpha D + \frac{1}{4}\left[1+\cos(2\lambda-2\varpi)\right]\alpha^2 D^2\right\}\cos[j(\lambda-\lambda')] \\
&= \frac{1}{8}\left(4\alpha D + 2\alpha^2 D^2\right)\cos[j(\lambda-\lambda')] \qquad (266)\\
&+ \frac{1}{8}\left(-2\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \\
&+ \frac{1}{8}\left(-2\alpha D + \alpha^2 D^2\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]
\end{aligned}$$

$$\begin{aligned}
G_1\cos[j(\theta-\theta')]_{e^2} &= \frac{1}{8}\left(4j^2-5j\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi] \qquad (267)\\
&+ \frac{1}{8}\left(4j^2+5j\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]
\end{aligned}$$

$$\begin{aligned}
G_e\cos[j(\theta-\theta')]_e &= \left[-\cos(\lambda-\varpi)\alpha D\right]\left\{j\cos[(1+j)\lambda - j\lambda' - \varpi] - j\cos[(1-j)\lambda + j\lambda' - \varpi]\right\} \\
&= \frac{1}{8}(-4j\alpha D)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \qquad (268)\\
&+ \frac{1}{8}(-4j\alpha D)\cos[j(\lambda - j\lambda')] \\
&+ \frac{1}{8}(4j\alpha D)\cos[(2-j)\lambda + j\lambda' - \varpi)] \\
&+ \frac{1}{8}(4j\alpha D)\cos[j(\lambda - j\lambda')] \\
&= \frac{1}{8}(-4j\alpha D)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \qquad (269)\\
&+ \frac{1}{8}(4j\alpha D)\cos[(2-j)\lambda + j\lambda' - \varpi)]
\end{aligned}$$

$$\begin{aligned}
\therefore \left\{G\cos[j(\theta-\theta')]\right\}_{e^2} &= \frac{1}{8}\left(4\alpha D + 2\alpha^2 D^2\right)\cos[j(\lambda-\lambda')] \qquad (270)\\
&+ \frac{1}{8}\left(5j + 4j^2 - 2\alpha D - 4j\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \\
&+ \frac{1}{8}\left(-5j + 4j^2 - 2\alpha D + 4j\alpha D + \alpha^2 D^2\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]
\end{aligned}$$

Order e'^2:

$$\left\{G\cos[j(\theta-\theta')]\right\}_{e'^2} = G_{e'^2}\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_{e'^2} + G_{e'}\cos[j(\theta-\theta')]_{e'}. \tag{271}$$

$$\begin{aligned}
& G_{e'^2}\cos[j(\theta-\theta')]_1 && (272)\\
= \;& \left\{\cos(2\lambda'-2\omega') + \left[\frac{1}{2}+\frac{3}{2}\cos(2\lambda'-2\varpi')\right]\alpha D + \frac{1}{4}[1+\cos(2\lambda'-2\varpi')]\alpha^2D^2\right\}\cos[j(\lambda-\lambda')]\\
= \;& \frac{1}{8}\left(4\alpha D + 2\alpha^2D^2\right)\cos[j(\lambda-\lambda')] && (273)\\
+ \;& \frac{1}{8}\left(4+6\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
+ \;& \frac{1}{8}\left(4+6\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

$$\begin{aligned}
G_1\cos[j(\theta-\theta')]_{e'^2} &= -j^2\cos[j(\lambda-\lambda')] && (274)\\
&+ \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']\\
&= \frac{1}{8}\left(-8j^2\right)\cos[j(\lambda-\lambda')] && (275)\\
&+ \frac{1}{8}\left(4j^2-5j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ \frac{1}{8}\left(4j^2+5j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

$$\begin{aligned}
& G_{e'}\cos[j(\theta-\theta)]e' && (276)\\
= \;& \cos(\lambda'-\varpi')(1+\alpha D)\left\{j\cos[(1+j)\lambda' - j\lambda - \varpi'] - j\cos[(1-j)\lambda' + j\lambda - \varpi']\right\}\\
= \;& \frac{1}{8}(4j+4j\alpha D)\cos[(2+j)\lambda' - j\lambda - 2\varpi'] + \frac{1}{8}(4j+4j\alpha D)\cos[j(\lambda-\lambda')] && (277)\\
+ \;& \frac{1}{8}(-4j-4j\alpha D)\cos[(2-j)\lambda' + j\lambda - 2\varpi'] + \frac{1}{8}(-4j-4j\alpha D)\cos[(j(\lambda-\lambda')]\\
= \;& \frac{1}{8}(4j+4j\alpha D)\cos[(2+j)\lambda' - j\lambda - 2\varpi'] && (278)\\
+ \;& \frac{1}{8}(-4j-4j\alpha D)\cos[(2-j)\lambda' + j\lambda - 2\varpi']
\end{aligned}$$

$$\begin{aligned}
\therefore \left\{G\cos[j(\theta-\theta')]\right\}_{e'^2} &= \frac{1}{8}\left(-8j^2+4\alpha D+2\alpha^2D^2\right)\cos[j(\lambda-\lambda')] && (279)\\
&+ \frac{1}{8}\left(4-9j+4j^2+6\alpha D-4j\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ \frac{1}{8}\left(4+9j+4j^2+6\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

Order ss':

$$\begin{aligned}
\left\{\delta\alpha\cos[j(\theta-\theta')]\right\}_{ss'} &= \delta_{ss'}\alpha\cos[j(\theta-\theta')]_1 && (280)\\
&= 2\alpha\left[\cos(\lambda-\lambda'-\Omega+\Omega')-\cos(\lambda+\lambda'-\Omega-\Omega')\right]\cos[j(\lambda-\lambda')] && (281)\\
&= \alpha\cos[(1-j)\lambda-(1-j)\lambda'-\Omega+\Omega'] && (282)\\
&+ \alpha\cos[(1+j)\lambda-(1+j)\lambda'-\Omega+\Omega']\\
&- \alpha\cos[(1-j)\lambda+(1+j)\lambda'-\Omega-\Omega']\\
&- \alpha\cos[(1+j)\lambda+(1-j)\lambda'-\Omega-\Omega']
\end{aligned}$$

Order s^2:

$$\begin{aligned}
\left\{\delta\alpha\cos[j(\theta-\theta')]\right\}_{s^2} &= \delta_{s^2}\alpha\cos[j(\theta-\theta')]_1 && (283)\\
&= \alpha[\cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda')]\cos[j(\lambda-\lambda')] && (284)\\
&= \frac{\alpha}{2}\cos[(1+j)\lambda+(1-j)\lambda'-2\Omega] && (285)\\
&+ \frac{\alpha}{2}\cos[(1-j)\lambda+(1+j)\lambda'-2\Omega]\\
&- \frac{\alpha}{2}\cos[(1+j)\lambda-(1+j)\lambda']\\
&- \frac{\alpha}{2}\cos[(1-j)\lambda+(1-j)\lambda'].
\end{aligned}$$

Order s'^2:

$$\begin{aligned}
\left\{\delta\alpha\cos[j(\theta-\theta')]\right\}_{s'^2} &= \delta_{s'^2}\alpha\cos[j(\theta-\theta')]_1 && (286)\\
&= \alpha[\cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda')]\cos[j(\lambda-\lambda')] && (287)\\
&= \frac{\alpha}{2}\cos[(1+j)\lambda'+(1-j)\lambda-2\Omega'] && (288)\\
&+ \frac{\alpha}{2}\cos[(1-j)\lambda'+(1+j)\lambda-2\Omega']\\
&- \frac{\alpha}{2}\cos[(1+j)\lambda-(1+j)\lambda']\\
&- \frac{\alpha}{2}\cos[(1-j)\lambda+(1-j)\lambda'].
\end{aligned}$$

If we now expand (42)

$$\mathcal{S}_D = \sum_{j=-\infty}^{\infty} \mathcal{S}_D^{(j)}, \tag{289}$$

we can express the components of individual terms:

$$\left(\mathcal{S}_D^{(j)}\right)_1 = \frac{1}{2}\cos[j(\lambda-\lambda')], \tag{290}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_e &= (1/8)\,(4j-2\alpha D)\cos((1+j)\lambda - j\lambda' - \varpi) \\ &+ (1/8)\,(-4j-2\alpha D)\cos((1-j)\lambda + j\lambda' - \varpi)\end{aligned} \tag{291}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{e'} &= (1/8)\,(2+4j+2\alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi'] \\ &+ (1/8)\,(2-4j+2\alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi']\end{aligned} \tag{292}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{ee'} &= (1/8)\left(2j+4j^2-2\alpha D-\alpha^2D^2\right)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)\left(-2j+4j^2-2\alpha D-\alpha^2D^2\right)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)\left(2j-4j^2-2\alpha D+4j\alpha D-\alpha^2D^2\right)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)\left(-2j-4j^2-2\alpha D-4j\alpha D-\alpha^2D^2\right)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'],\end{aligned} \tag{293}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{e^2} &= (1/8)\left(2\alpha D+\alpha^2D^2\right)\cos[j(\lambda-\lambda')] \\ &+ (1/16)\left(5j+4j^2-2\alpha D-4j\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda-j\lambda'-2\varpi] \\ &+ (1/16)\left(-5j+4j^2-2\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda+j\lambda'-2\varpi]\end{aligned} \tag{294}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{e'^2} &= (1/8)\left(-4j^2+2\alpha D+\alpha^2D^2\right)\cos[j(\lambda-\lambda')] \\ &+ (1/16)\left(4-9j+4j^2+6\alpha D-4j\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda'+j\lambda-2\varpi'] \\ &+ (1/16)\left(4+9j+4j^2+6\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda'-j\lambda-2\varpi']\end{aligned} \tag{295}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{ss'} &= (1/2)\alpha\cos[(1-j)\lambda-(1-j)\lambda'-\Omega+\Omega'] \\ &+ (1/2)\alpha\cos[(1+j)\lambda-(1+j)\lambda'-\Omega+\Omega'] \\ &- (1/2)\alpha\cos[(1-j)\lambda+(1+j)\lambda'-\Omega-\Omega'] \\ &- (1/2)\alpha\cos[(1+j)\lambda+(1-j)\lambda'-\Omega-\Omega']\end{aligned} \tag{296}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{s^2} &= (1/4)\alpha\cos[(1+j)\lambda+(1-j)\lambda'-2\Omega] \\ &+ (1/4)\alpha\cos[(1-j)\lambda+(1+j)\lambda'-2\Omega] \\ &- (1/4)\alpha\cos[(1+j)\lambda-(1+j)\lambda'] \\ &- (1/4)\alpha\cos[(1-j)\lambda+(1-j)\lambda'],\end{aligned} \tag{297}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{s'^2} &= (1/4)\alpha\cos[(1+j)\lambda'+(1-j)\lambda-2\Omega'] \\ &+ (1/4)\alpha\cos[(1-j)\lambda'+(1+j)\lambda-2\Omega'] \\ &- (1/4)\alpha\cos[(1+j)\lambda-(1+j)\lambda'] \\ &- (1/4)\alpha\cos[(1-j)\lambda+(1-j)\lambda'].\end{aligned} \tag{298}$$

A Series Expansions

In this section, we develop polynomial approximations for E, $\sin E$, $\cos E$, $\sin f$, $\cos f$ in terms of the mean anomaly M and eccentricity e. First, series expansions will be derived as Fourier series expansion w.r.t. M with coefficients that involve Bessel functions and eccentricity. Then, using polynomial approximation for the Bessel functions, the expressions will be expanded to a finite order of e.

Any sufficiently well-behaved function $g(M)$ defined on the interval $[0, 2\pi]$ can be expanded as a **Fourier series**

$$g(M) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_n \cos kM \; dx + \sum_{k=1}^{\infty} b_n \sin kM \; dx, \tag{299}$$

where the Fourier series coefficients are defined

$$a_0 := \frac{1}{2\pi} \int_0^{2\pi} g(M) \, dM, \tag{300}$$

$$a_k := \frac{1}{\pi} \int_0^{2\pi} g(M) \cos kM \; dM, \tag{301}$$

$$b_k := \frac{1}{\pi} \int_0^{2\pi} g(M) \sin kM \; dM, \tag{302}$$

Bessel functions of the first kind can be defined

$$J_k(x) := \frac{1}{2\pi} \int_0^{2\pi} \cos(kt - x \sin t) \, dt \tag{303}$$

We also will use the following recurrence relations

$$J_k(x) = \frac{x}{2k}[J_{k-1}(x) + J_{k+1}(x)], \tag{304}$$

$$J_k'(x) = \frac{1}{2}[J_{k-1}(x) - J_{k+1}(x)] \tag{305}$$

and the following approximations

$$J_1(x) = \frac{1}{2}x - \frac{1}{16}x^3 + \frac{1}{384}x^5 + \mathcal{O}(x^7), \tag{306}$$

$$J_2(x) = \frac{1}{8}x^2 - \frac{1}{96}x^4 + \mathcal{O}(x^6), \tag{307}$$

$$J_3(x) = \frac{1}{48}x^3 - \frac{1}{768}x^5 + \mathcal{O}(x^7), \tag{308}$$

$$J_4(x) = \frac{1}{384}x^4 + \mathcal{O}(x^6), \tag{309}$$

$$J_5(x) = \frac{1}{3840}x^5 + \mathcal{O}(x^7). \tag{310}$$

A.1 Eccentric Anomaly and True Anomaly

A.1.1 E

Let's try to expand $g(M) := e \sin E(M)$ as a Fourier series w.r.t. M, where the relationship between the mean and eccentric anomalies follows from the Kepler Equation $M = E - e \sin E$. Then,

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} e \sin[E(M)]\, dM. \tag{311}$$

Change of integration variable from M to E does not change the limits from $dM = (1 - e \cos E)\, dE$, we obtain

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} e \sin E(1 - e \cos E)\, dE = \frac{e}{2\pi} \int_0^{2\pi} \sin E\, dE - \frac{e^2}{2\pi} \int_0^{2\pi} \sin E \cos E\, dE = 0. \tag{312}$$

Since $\sin E(M)$ is anti-symmetric and $\cos kM$ is symmetric, we also get

$$a_k = \frac{1}{\pi} \int_0^{2\pi} e \sin\,[E(M)] \cos kM\, dM = 0. \tag{313}$$

On the other hand,

$$b_k = \frac{1}{\pi} \int_0^{2\pi} e \sin[E(M)] \sin kM\, dM \tag{314}$$

$$= \frac{1}{\pi} \int_0^{2\pi} e \sin E \sin\,[k(E - e \sin E)]\,(1 - e \cos E)\, dE \tag{315}$$

We apply integration by parts $\int u\, dv = uv - \int v\, du$, where $u = e \sin E$ and

$$dv = \sin[k(E - e \sin E)](1 - e \cos E)\, dE = -\left\{ \frac{1}{k} \frac{d}{dE} \cos[k(E - e \sin E)] \right\} dE. \tag{316}$$

Thus, $v = -\frac{1}{k} \cos[k(E - e \sin E)]$

$$b_k = -\Big|_0^{2\pi} \frac{e}{k\pi} \sin E \cos[k(E - e \sin E)] + \frac{e}{k\pi} \int_0^{2\pi} \cos[k(E - e \sin E) \cos E\, dE. \tag{317}$$

First term obviously disappears and for the second term, we can apply the trigonometric formula $\cos x \cos y = \frac{1}{2} \cos(x + y) + \frac{1}{2} \cos(x - y)$ to obtain

$$b_k = \frac{e}{2k\pi} \int_0^{2\pi} \cos[(k + 1)E - ke \sin E]\, dE + \frac{e}{2k\pi} \int_0^{2\pi} \cos[(k - 1)E - ke \sin E]\, dE. \tag{318}$$

Application of (303) and (304), yields

$$J_n(ke) = \frac{1}{2\pi} \int_0^{2\pi} \cos(nE - ke \sin E)\, dE. \tag{319}$$

and

$$b_k = \frac{e}{k}\left[J_{k+1}(ke) + J_{k-1}(ke)\right] = \frac{2}{k}J_k(ke), \tag{320}$$

Therefore, we obtain the series expansion

$$E = M + e\sin E = M + \sum_{k=1}^{\infty}\frac{2J_k(ke)}{k}\sin kM. \tag{321}$$

Expanding the individual terms

$$\begin{aligned}
2J_1(e)\sin M &\approx \left(e - \frac{1}{8}e^3 + \frac{1}{192}e^5 + \mathcal{O}(e^7)\right)\sin M && (322)\\
J_2(2e)\sin 2M &\approx \left(\frac{1}{2}e^2 - \frac{1}{6}e^4 + \mathcal{O}(e^6)\right)\sin 2M && (323)\\
\frac{2}{3}J_3(3e)\sin 3M &\approx \left(\frac{3}{8}e^3 - \frac{27}{128}e^5 + \mathcal{O}(e^7)\right)\sin 3M && (324)\\
\frac{1}{2}J_4(4e)\sin 4M &\approx \left(\frac{1}{3}e^4 + \mathcal{O}(e^6)\right)\sin 4M && (325)
\end{aligned}$$

Thus, we finally obtain the approximation

$$\begin{aligned}
E &\approx M + e\sin M + \frac{1}{2}e^2\sin 2M + e^3\left(-\frac{1}{8}\sin M + \frac{3}{8}\sin 3M\right) && (326)\\
&+ e^4\left(-\frac{1}{6}\sin 2M + \frac{1}{3}\sin 4M\right) + \mathcal{O}(e^5).
\end{aligned}$$

A.1.2 $\cos nE$

Let now $g(M) := \cos[nE(M)]$. Then, clearly

$$\begin{aligned}
a_0 &= \frac{1}{2\pi}\int_0^{2\pi}\cos nE\, dM && (327)\\
&= \frac{1}{2\pi}\int_0^{2\pi}\cos nE(1 - e\cos E)dE && (328)\\
&= \frac{1}{2\pi}\int_0^{2\pi}\cos nE\, dE - e\frac{1}{2\pi}\int_0^{2\pi}\cos nE\cos E\, dE = -\frac{e}{2\pi}\frac{2\pi}{2}\delta_{n1} = -\frac{e}{2}\delta_{n1}, && (329)\\
b_k &= \frac{1}{\pi}\int_0^{2\pi}\sin kM\cos nE\, dM = 0 && (330)
\end{aligned}$$

and by taking into account that $\cos kM = (1/k)d(\sin kM)/dM$, we also obtain

$$\begin{aligned}
a_k &= \frac{1}{\pi}\int_0^{2\pi}\cos kM\cos nE\, dM && (331)\\
&= \frac{1}{k\pi}\Big|_0^{2\pi}\cos nE\sin kM - \frac{1}{k\pi}\int_0^{2\pi}\frac{d\cos nE}{dM}\sin kM\, dM && (332)
\end{aligned}$$

We perform a change of variables from M to E by noting that

$$\sin kM = \sin[k(E - e\sin E)], \tag{333}$$

$$\frac{d(\cos nE)}{dM} = -n\sin nE\frac{dE}{dM}. \tag{334}$$

Thereafter, substitution $\cos(x-y) - \cos(x+y) = 2\sin x \sin y$, yields

$$a_k = \frac{n}{k\pi}\int_0^{2\pi} \sin nE \sin[k(E - e\sin E)]\, dE \tag{335}$$

$$= \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k-n)E - ke\sin E)]\, dE - \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k+n)E - ke\sin E)]\, dE \tag{336}$$

$$= \frac{n}{k}\left[J_{k-n}(ke) - J_{k+n}(ke)\right] \tag{337}$$

and finally

$$\cos nE = -\frac{e}{2}\delta_{n1} + \sum_{k=1}^{\infty}\frac{n}{k}[J_{k-n}(ke) - J_{k+n}(ke)]\cos kM. \tag{338}$$

In the important special case $n = 1$

$$\cos E = -\frac{e}{2} + \sum_{k=1}^{\infty}\frac{2}{k}\frac{dJ_k}{de}(ke)\cos kM. \tag{339}$$

A.1.3 $\sin nE$

Let $g(M) = \sin[nE(M)]$. Then,

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} \sin nE\, dM \tag{340}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \sin nE(1 - e\cos E)\, dE \tag{341}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \sin nE\, dE - \frac{e}{2\pi}\int_0^{2\pi} \sin nE\cos E\, dE = 0. \tag{342}$$

$$a_k = \frac{1}{\pi}\int_0^{2\pi} \sin nE\cos kM\, dM = 0. \tag{343}$$

Using $\sin kM = -(1/k)(d/dM)\cos kM$, we can compute

$$b_k = \frac{1}{\pi}\int_0^{2\pi} \sin nE\sin kM\, dM \tag{344}$$

$$= -\frac{1}{k\pi}\Big|_0^{2\pi} \sin nE\cos kM + \frac{1}{k\pi}\int_0^{2\pi}\frac{d(\sin nE)}{dM}\cos kM\, dM \tag{345}$$

$$= \frac{n}{k\pi}\int_0^{2\pi} \cos nE\frac{dE}{dM}\cos kM\, dM \tag{346}$$

$$= \frac{n}{k\pi}\int_0^{2\pi} \cos nE\cos k(E - e\sin E)\, dE. \tag{347}$$

Using $\cos x \cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$, we obtain

$$b_k = \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k+n)E - ke\sin E)]\,dE + \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k-n)E - ke\sin E)]\,dE \quad (348)$$

$$= \frac{n}{k}\left[J_{k+n}(ke) - J_{k-n}(ke)\right] \quad (349)$$

Thus, we finally obtain

$$\sin nE = \sum_{k=1}^{\infty} \frac{n}{k}\left[J_{k+n}(ke) - J_{k-n}(ke)\right]\sin kM \quad (350)$$

and in the special case $n = 1$

$$\sin E = \sum_{k=1}^{\infty} \frac{2}{k}\frac{dJ_k}{de}(ke)\sin kM \quad (351)$$

A.1.4 $\cos f$

A.1.5 $\sin f$

A.2 r/a

References

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[2] Murray, Dermott, Solar System Dynamics, Cambridge University Press, 1999